31 Priority Hackathon Problem Statements

CGC UNIVERSITY, MOHALI

AI | AGENTIC SYSTEMS | REAL-WORLD IMPACT

Selection emphasizes impact, AI relevance, originality, feasibility and demo readiness within 24 hours.

01. AgentSphere: Collaborative AI Agents for Complex Tasks

Create a multi-agent system in which specialized agents collaborate to complete a defined task, such as researching a topic, validating sources, analysing data, and generating a final report with traceable evidence.

AI: Agentic AI, tool use, workflow orchestration, RAG

02. PhishShield AI: Phishing and Scam Detection

Build a browser extension or web application that analyses emails, SMS messages, URLs, and message content to flag likely phishing attempts, suspicious links, impersonation, and social-engineering patterns.

AI: NLP, URL classification, anomaly detection

03. Real-Time UPI Social Engineering Phishing Interceptor

The Challenge: A low-latency on-device utility that analyzes patterns in high-pressure phishing phone language, suspicious deep-link clicks, and remote access apps.

Bharat Context: Safely freezes accidental UPI or digital wallet transactions right before authorization to protect digital-illiterate users from scams.

SDG Alignment: SDG 16: Peace, Justice, and Strong Institutions

04. DeepTruth: AI-Based Deepfake Detection

Develop a tool that analyses images, audio clips, or videos for manipulation indicators and provides a confidence score with supporting evidence and limitations.

AI: Computer vision, audio analysis, multimodal learning

05. RescueAI: Disaster Risk and Resource Coordination

Build a decision-support platform that combines weather alerts, geographic data, and reported incidents to map potential risk areas, prioritize requests, and recommend resource distribution.

AI: Geospatial analytics, forecasting, resource optimization

06. Crowdsourced Extreme Weather Verification & GIS De-Duplicator

The Challenge: An AI dashboard that ingests chaotic crowd-sourced public multimedia reports from social media during active disasters, filters out old fake media duplicates, and verifies locations.

Bharat Context: Maps verified reports instantly into an India Meteorological Department (IMD) or NDMA disaster control grid.

SDG Alignment: SDG 13: Climate Action

07. KrishiVision AI: Crop Disease Detection

Create a mobile application that analyses crop-leaf images to identify possible plant diseases and provides preventive guidance, including uncertainty warnings when the image is unclear.

AI: Computer vision, image classification, explainable AI

08. Automated Crop-Disease Scanner on Low-Cost Hardware

The Challenge: An offline-first mobile app using edge computer vision (TensorFlow Lite) to scan local Indian staple crops (mustard, cotton, rice) for pests.

Bharat Context: Must run entirely on sub-₹10,000 Android phones without requiring active cloud connectivity.

SDG Alignment: SDG 15: Life on Land

09. BhashaBridge: Multilingual AI Voice Assistant

Create a voice-based interface that understands Indian languages and mixed-language speech, translates queries, and provides spoken access to digital services for users with limited English proficiency.

AI: Speech recognition, translation, multilingual NLP

10. Indic Voice-First Agri-Advisory Bot for Mandi Pricing

The Challenge: A regional-language, voice-driven AI agent that handles mixed dialects (like Hinglish or Tamil-English) to provide farmers with microclimate insights and crop pricing.

Bharat Context: Designed for semi-literate farmers using low-bandwidth 2G/3G networks.

SDG Alignment: SDG 1: No Poverty & SDG 2: Zero Hunger

11. CitizenAssist AI: Intelligent Public Service Navigator

Create a multilingual assistant that helps citizens locate relevant public services, understand eligibility criteria, prepare document checklists, and find official application links.

AI: RAG, multilingual NLP, information extraction

12. FoodLink AI: Surplus Food Redistribution Platform

Develop a system that predicts surplus food availability, matches donors with nearby recipient organizations, and recommends collection routes based on urgency, distance, and food safety constraints.

AI: Matching, demand forecasting, route optimization

13. NaviAble AI: Assistive Navigation for Visually Impaired Users

Build a smartphone-based assistant that identifies nearby obstacles and common objects, reads signboards aloud, and provides contextual navigation assistance.

AI: Object detection, OCR, text-to-speech, edge AI

14. ASHA Worker Voice-to-EHR Multi-Modal Co-Pilot

The Challenge: A tablet-based AI helper for rural ASHA health workers that converts unstructured audio summaries recorded in native dialects into structured, digitally mapped electronic health records (EHR).

Bharat Context: Handles heavy code-switching (e.g., Marathi keywords mixed into spoken Hindi).

SDG Alignment: SDG 3: Good Health and Well-Being

15. AccessLearn AI: Accessible Classroom Learning

Develop a tool that converts lectures into captions, searchable notes, simplified summaries, and multilingual transcripts, with adjustable reading levels for different learners.

AI: Speech recognition, summarization, translation

16. CareerMatch AI: Resume and Job Skill Matching

Develop a platform that compares student resumes with job descriptions, identifies missing competencies, and recommends personalized learning pathways and projects.

AI: NLP, semantic matching, recommendation systems

17. InterviewIQ: AI Mock Interview Coach

Create an interview simulator that asks role-specific questions, evaluates the relevance of answers, identifies gaps in technical knowledge, and gives structured feedback on communication.

AI: LLMs, speech processing, rubric-based assessment

18. DocuMind: Intelligent Document Processing System

Develop a system that extracts structured information from invoices, forms, certificates, and reports, identifies missing fields, detects inconsistencies, and generates summaries automatically.

AI: OCR, computer vision, document LLMs

19. AutoFlow AI: Intelligent Workflow Automation

Create an AI system that converts natural-language instructions into approval workflows, assigns tasks, identifies bottlenecks, and recommends automation for repetitive administrative processes.

AI: LLM agents, process mining, predictive analytics

20. CodeGuard AI: Intelligent Code Review Assistant

Develop a coding assistant that identifies bugs, insecure coding patterns, duplicated logic, and performance bottlenecks while suggesting corrections and explaining the reasoning behind them.

AI: Code LLMs, static analysis, vulnerability detection

21. VerifyAI: Misinformation and Claim Verification Assistant

Create a system that extracts factual claims from online content, retrieves relevant credible sources, compares evidence, and presents a transparent assessment rather than relying on unsupported AI-generated verdicts.

AI: NLP, semantic search, RAG, evidence ranking

22. WasteWise AI: Automated Waste Segregation

Build a computer-vision prototype that classifies waste into categories such as plastic, paper, metal, glass, and organic waste to support recycling and efficient waste handling.

AI: Image classification, object detection, edge AI

23. Public Transit Pothole & Road Anomaly Mapping Engine

The Challenge: A dashcam-based computer vision architecture that aggregates live feeds from state buses or public utility vehicles to automatically geo-tag and rank road deterioration levels.

Bharat Context: Gives Municipal Corporations a low-cost, automated way to map rural road networks without dedicated inspection vehicles.

SDG Alignment: SDG 9: Industry, Innovation, and Infrastructure

24. Dynamic Crowdsourced Waste Dumping Detection & Routing

The Challenge: An AI system that processes public images of illegal dumping sites, categorizes the waste volume using an automated 'Waste Quantity Index', and dynamically optimizes route planning for municipal garbage trucks.

Bharat Context: Adapts to highly unstructured, chaotic garbage dumping patterns in Tier-2 Indian cities.

SDG Alignment: SDG 11: Sustainable Cities and Communities & SDG 12: Responsible Consumption and Production

25. Vernacular OCR Hand-written Invoice-to-GST Formatter

The Challenge: A mobile app enabling small tier-3 shopkeepers to take a picture of a handwritten bill (Kacha Bill) and automatically format it into structured, tax-compliant data for GST filing portals.

Bharat Context: Understands non-standard shorthand scripts and regional counting notations.

SDG Alignment: SDG 8: Decent Work and Economic Growth

26. AquaSense AI: Intelligent Water Leakage Detection

Build an IoT-based solution that uses flow, pressure, and consumption readings to detect abnormal water usage, identify probable leaks, and prioritize maintenance.

AI: Time-series anomaly detection, IoT analytics

27. IrrigaSmart: AI-Based Precision Irrigation

Develop an intelligent irrigation assistant that estimates crop water requirements from soil moisture, weather, crop type, and growth-stage data to recommend irrigation schedules.

AI: Predictive modeling, sensor analytics, optimization

28. Automated Cyber-Bullying & Toxic Content Filter for Indic Languages

The Challenge: Develop an LLM-based regional content monitoring tool to flag cyber-bullying, localized hate speech, and harassment across multi-lingual student community channels.

Bharat Context: Accurately processes localized slangs, phonetic misspellings, and Romanized script writing.

SDG Alignment: SDG 4: Quality Education & SDG 16: Peace, Justice, and Strong Institutions

29. Early Forest Fire & Heat Anomaly Detection via Edge-AI Drones

Theme 6: Disaster Management & Climate Anomaly Analytics

The Challenge: A computer vision payload optimized for cheap commercial drones that performs real-time edge processing to flag thermal anomalies, microclimate smoke trails, and forest fire vectors before they spread.

Bharat Context: Built specifically for dense, unmapped forest covers in Western Ghats or North-East regions with zero cellular network backup.

SDG Alignment: SDG 13: Climate Action & SDG 15: Life on Land

30. Indian Sign Language (ISL) Interactive Classroom Interpreter

The Challenge: A real-time computer vision interpreter that translates a government school teacher's live audio lecture into continuous Indian Sign Language (ISL) animations on a screen.

Bharat Context: Built using low-compute models that run on simple smart classroom TVs without high-end GPUs.

SDG Alignment: SDG 4: Quality Education & SDG 10: Reduced Inequalities

31. Open Innovation

Theme: Open Innovation | Any Domain

The Challenge: Identify a real-world problem in any domain and develop an original, practical solution. Teams should define the problem, identify intended users, explain their approach, and demonstrate a feasible prototype with clear potential for impact.

Evaluation Focus: Originality, usefulness, feasibility, implementation, and potential impact.